



POLITECNICO
MILANO 1863

Final Project Econometrics

Group 9

Francesco Biancini

`francesco.biancini@mail.polimi.it`

Davide Matteo Bottinelli

`davidematteo.bottinelli@mail.polimi.it`

Salvatore Ippolito

`salvatore1.ippolito@mail.polimi.it`

Mario Pennino

`mario.pennino@mail.polimi.it`

Contents

1	Project Overview	1
2	Dataset preparation	1
2.1	Data Sources & Variables	1
3	Data Mining and Feature Engineering	1
3.1	Logarithmic Transformations	1
3.2	Unit Root Testing	2
3.3	Multicollinearity	2
4	Regressions models	2
4.1	Assumptions	2
4.2	Random and Fixed Effects	4
4.3	LME model	5
5	Local Projections Analysis	5
5.1	Mathematical Framework and Rationale	6
5.2	Empirical Results and Discussion	6
6	Panel VAR Analysis	6
6.1	Mathematical Framework and Identification	6
6.2	Lag Selection and Methodological Advantages	7
6.3	Empirical Results and Economic Discussion	7
7	ARMA Models	8
7.1	Basic ARMA	8
7.2	ARMAX Extension	9
8	Forecasting	9
8.1	Forecasting Scenarios	10
9	Excluded models	10

1 Project Overview

The primary objective of this research is to quantify the dynamic relationship between granular climate shocks and aggregate macroeconomic performance. Unlike traditional studies that rely on mean temperature shifts, this project focuses on the frequency of climate extremes—specifically heatwaves, severe cold spells, and extreme precipitation events.

The empirical strategy utilizes a highly balanced panel dataset of countries over a 63-year historical period (1960–2023). To ensure robust causal inference, the methodology addresses key econometric challenges inherent in macroeconomic panel data, including non-stationarity, cross-sectional dependence, heteroskedasticity, and outlier distortion. After establishing the historical baseline, the study projects these economic impacts to 2050 under different Shared Socioeconomic Pathways (SSPs).

2 Dataset preparation

2.1 Data Sources & Variables

Table following table provides a summary of the core variables utilized in this analysis.

Variable	Description	Source
GDP	Real Gross Domestic Product (rgdpna)	PWT 11.0
emp	Total Employment	PWT 11.0
rnna	Capital Stock at constant national prices	PWT 11.0
hc	Human Capital Index (based on schooling years)	PWT 11.0
infl	Annual Inflation Rate	GMD
hd30	Number of days with maximum temperature $> 30^{\circ}\text{C}$	World Bank climate knowledge portal
cdd	Cold Degree Days	World Bank climate knowledge portal
r20mm	Number of days with precipitation $> 20\text{mm}$	World Bank climate knowledge portal

Macroeconomic data frequently contains systemic shocks (e.g., hyperinflation, war, state collapse) that do not reflect normal economic dynamics. To systematically identify these observations, an auxiliary Least Squares Dummy Variable (LSDV) regression was specified. Influence diagnostics were extracted, specifically focusing on Leverage, Studentized Residuals, and Cook’s Distance. Severe distortions were observed from Venezuela (driven by hyperinflation and economic collapse), Iran, and Algeria, whose residuals frequently exceeded ± 3 standard deviations. These three countries were permanently excluded from the base dataset.

3 Data Mining and Feature Engineering

Raw data was mathematically transformed to stabilize variances, satisfy linear regression assumptions, and isolate exogenous climate signals from predictable seasonal trends.

3.1 Logarithmic Transformations

To interpret regression coefficients as elasticities (percentage changes), the natural logarithm was applied to highly skewed macroeconomic variables: GDP, employment (**emp**), and capital stock (**rnna**). Because inflation (**infl**) frequently contains negative values, a sign-preserving logarithmic transformation was applied:

$$\log_infl = \text{sign}(infl) \times \ln(1 + |infl|) \quad (1)$$

3.2 Unit Root Testing

Two primary tests were employed: Country-specific Augmented Dickey-Fuller (ADF) tests and panel-wide Im-Pesaran-Shin (IPS) tests^[4].

- **Macroeconomic Variables:** Log-GDP, log-employment, log-capital stock, and human capital exhibited a clear deterministic trend and failed the unit root tests in levels. They were classified as $I(1)$. To achieve stationarity, these variables were transformed into their first differences (e.g., d_log_GDP). Subsequent ADF testing confirmed these differenced variables are actually $I(0)$.
- **Climate Shock Variables:** Thanks to the expanding window methodology, the climate shocks mean-revert around zero. IPS testing yielded p -values < 0.001 , confirming that $hd30$, cdd , and $r20mm$ are $I(0)$.

3.3 Multicollinearity

To ensure the independence of the regressors, a Pearson correlation matrix was generated, revealing no severe linear relationships among the exogenous variables. Furthermore, the Variance Inflation Factor (VIF) was computed for each covariate; the resulting values were found to be approximately equal to 1, confirming the absence of multicollinearity.

4 Regressions models

We built a sequence of panel specifications to study the relationship between GDP growth and climate-related covariates. The starting point was a baseline panel regression, and the model selected to test the assumptions was the fixed-effects specification, estimated with country and year effects. This choice was driven by the structure of the data and by the Hausman test, which indicated that the random-effects model was not appropriate for the main interpretation.

The dependent variable is d_log_GDP , the first difference of log GDP, so the coefficients can be interpreted as semi-elasticities. The baseline specification is

$$\Delta \log(GDP)_{it} = \beta_1 hd30_{it} + \beta_2 r20mm_{it} + \beta_3 cdd_{it} + \beta_4 \Delta \log(emp)_{it} + \beta_5 \Delta hc_{it} + \beta_6 \log(infl)_{it} + \beta_7 \Delta \log(rnna)_{it} + u_{it}.$$

4.1 Assumptions

The first assumption is the zero-mean condition, $E[u_{it}] = 0$. Because the true disturbances are not observed, the relevant diagnostic is the mean of the estimated residuals. In the fixed-effects model this mean is

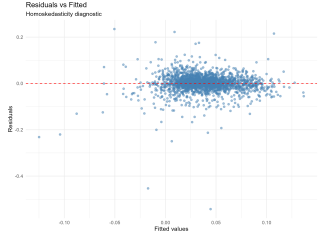
$$\bar{\hat{u}} = 3.72 \times 10^{-19},$$

which is effectively zero. This is expected because OLS chooses the fitted coefficients so as to minimize the residual sum of squares, which implies that the average residual is zero in the presence of a constant term or an equivalent fixed-effects centering.

The second assumption is homoskedasticity. In this project the relevant diagnostic is White's test, which is based on an auxiliary regression for the squared residuals. The null hypothesis is homoskedasticity. If it is rejected, the coefficient estimates remain valid, but the usual standard errors may no longer be reliable.

The check is important here because the sample is an unbalanced macro-panel of 32 countries over 46–63 years, so the variance of shocks can easily differ across economies and over time. This is precisely the kind of setting where White's test is informative.

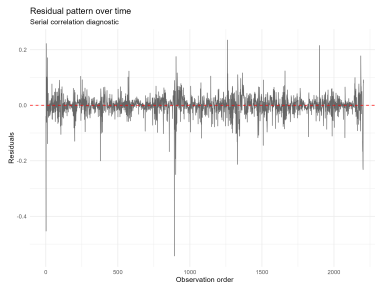
Statistic	Value
Chi-squared	89.682
Degrees of freedom	2
p-value	< 0.001



The White test strongly rejects the null hypothesis of homoskedasticity. This means that the variance of the disturbances is not constant across observations, so conventional standard errors may be unreliable.

The residual-versus-fitted plot should be read together with White's test. A visible fan shape or changing dispersion would indicate heteroskedasticity, while a roughly constant spread around zero would support the homoskedasticity assumption. Here, combining the available information, we conclude that heteroskedasticity is present (so we will account for it with LME and Driscoll-Kraay standard errors).

The third assumption is no serial correlation, $Cov[u_{it}, u_{is}] = 0$ for $t \neq s$. This is especially relevant for yearly panel data, because climate shocks and macroeconomic dynamics often persist over time. If residuals are serially correlated, standard errors can be understated and significance can be exaggerated.

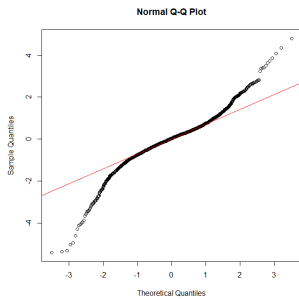


The residual pattern is mostly centered around zero and does not display a clear deterministic trend or a long-lasting smooth cycle. This suggests that the fixed-effects specification captures the main systematic component of GDP growth reasonably well. At the same time, the plot shows a few sharp spikes and short clusters of larger positive and negative residuals, which indicate occasional shocks and some local dependence over time. Overall, the graph does not point to severe serial correlation, but it does show that a few periods are characterized by stronger residual volatility.

The fourth assumption is exogeneity, meaning that the regressors should not be correlated with the disturbance term. This assumption cannot be tested directly, because the true errors are unobserved. For this reason, it is treated as a modelling assumption and assessed indirectly through the choice of specification and the economic plausibility of the regressors.

The fifth assumption is Gaussianity of the disturbances. This assumption is not essential for consistency, but it is important for classical inference and for assessing whether the residuals are well behaved. We first checked the residuals of the initial model using the Jarque-Bera test, which strongly rejected normality:

$$X^2 = 155686, \quad p\text{-value} < 2.2 \times 10^{-16}.$$



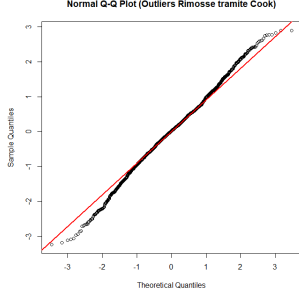
The corresponding Q-Q plot also showed substantial deviations from the reference line, confirming that the residuals exhibited heavy tails and the presence of extreme outliers. In particular, the departures observed in the tails indicate that the distribution of residuals differs markedly from the normal distribution, suggesting that a small number of observations exert a disproportionate influence on the overall error structure.

To improve the robustness of the specification, we then estimated an auxiliary model with country and year fixed effects and used Cook's distance to remove highly influential observations (this modified dataset will be

used only for non lagged models). After excluding the observations above the threshold $4/n$, we re-estimated the model and repeated the normality test on the cleaned residuals. The Jarque-Bera statistic dropped sharply to

$$X^2 = 7.9054, \quad p\text{-value} = 0.0192,$$

showing a much closer approximation to Gaussianity, although normality is still rejected at the 5% level.



This result suggests that the initial departure from normality was largely driven by a small number of influential observations. In the final Q–Q plot, the residuals lie much closer to the reference line, indicating a substantial improvement in the distributional properties of the model errors. The final residual distribution is therefore considerably more regular, and the cleaned specification provides a more reliable basis for statistical interpretation and inference.

4.2 Random and Fixed Effects

Table 1: Regression Results

	(1)	(2)	(3)	(4)
<i>hd30</i>	-0.000169*** (-2.707)	-0.000162*** (-2.744)	-0.000162*** (-2.666)	-0.000109 (-1.631)
<i>r20mm</i>	0.000069 (0.277)		0.000221 (1.215)	
<i>cdd</i>	0.000034 (0.563)		0.00000006 (0.001)	
$\Delta \log(emp)$	0.5021*** (6.199)	0.5036*** (6.099)	0.5048*** (6.232)	0.3347*** (11.770)
Δhc	0.0279 (0.610)		0.0258 (0.562)	0.0114 (0.332)
$\log(infl)$	-0.002241* (-1.951)	-0.002199* (-1.895)	-0.002244** (-1.990)	-0.001301* (-1.928)
$\Delta \log(rnna)$	0.5608*** (16.217)	0.5599*** (15.919)	0.5614*** (15.775)	0.5603*** (22.895)
<i>hd30</i> $\times$ <i>r20mm</i>			-0.0000054 (-0.548)	
<i>hd30</i> $\times$ <i>cdd</i>			0.0000034 (0.641)	
<i>r20mm</i> $\times$ <i>cdd</i>			0.0000344 (1.105)	
<i>hd30</i> ²				-0.000000567 (-0.433)
Adjusted R^2	0.367	0.367	0.366	0.309
Countries	32	32	32	32
Observations	1902	1902	1902	1902

Driscoll–Kraay (SCC) standard errors in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The empirical comparison between the two specifications is carried out with the Hausman test. In our case, the test rejects the null hypothesis that random effects are consistent, with a chi-squared statistic of 21.657 and a p-value of 0.00291. This result supports the use of the fixed-effects estimator as the baseline model.

To account for remaining cross-sectional and temporal dependence, the fixed-effects estimates are reported with Driscoll–Kraay standard errors. This correction is important in a panel with many countries and many years, because it produces inference that is robust to heteroskedasticity, autocorrelation and cross-sectional dependence.

4.3 LME model

Table 2: Linear Mixed-Effects Models

	(1)	(2)	(3)	(4)
<i>hd30</i>	-0.000142*** (-3.602)	-0.000135*** (-3.510)	-0.000139*** (-3.334)	-0.000076 (-1.219)
<i>r20mm</i>	0.000146 (0.736)		0.000245 (0.894)	
<i>cdd</i>	0.000018 (0.304)		-0.000016 (-0.216)	
$\Delta \log(emp)$	0.4971*** (18.246)	0.5005*** (18.414)	0.4993*** (18.258)	0.4947*** (18.093)
Δhc	0.0519 (1.312)		0.0502 (1.270)	0.0530 (1.345)
$\log(infl)$	-0.002229*** (-4.152)	-0.002146*** (-4.016)	-0.002224*** (-4.118)	-0.002219*** (-4.123)
$\Delta \log(rnna)$	0.5935*** (28.881)	0.5912*** (28.693)	0.5937*** (28.785)	0.5928*** (28.762)
<i>hd30</i> $\times$ <i>r20mm</i>			-0.000003 (-0.285)	
<i>hd30</i> $\times$ <i>cdd</i>			0.000004 (0.816)	
<i>r20mm</i> $\times$ <i>cdd</i>			0.000030 (0.899)	
<i>hd30</i> ²				-0.000002 (-1.330)
AIC	-8907.35	-8948.50	-8839.65	-8918.14
BIC	-8846.34	-8904.11	-8762.02	-8862.67

t-statistics in parentheses.

Random intercepts for countries and years.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

As a robustness check, we estimated a linear mixed-effects model using `nlme`. This specification relaxes the strict homoskedasticity and independence assumptions of the standard linear panel model by allowing for random intercepts at the country and year levels and by modeling heterogeneous variances through `varIdent()`. In this sense, the mixed-effects approach provides a more flexible alternative to the baseline fixed-effects regression.

The full specification includes the same covariates as the baseline model:

$$d_log_GDP_{it} = \mathbf{X}_{it}^\top \boldsymbol{\beta} + a_i + \lambda_t + \varepsilon_{it} \quad (2)$$

where a_i captures country-specific random heterogeneity and λ_t captures year-specific random variation.

The estimated mixed-effects model confirms the main results of the baseline regression. In the full specification, *hd30* is negative and significant, *d_log_emp* is strongly positive, *log_infl* is negative and significant, and *d_log_rnna* remains positive and highly significant. By contrast, *r20mm*, *cdd*, and *d_hc* are not statistically significant.

A parsimonious version of the model, which retains only *hd30*, *d_log_emp*, *log_infl*, and *d_log_rnna*, improves the information criteria, with a lower AIC and BIC than the full specification. This suggests that the main explanatory power is concentrated in these four regressors.

Overall, the mixed-effects model leads to the same qualitative conclusions as the baseline panel regression: heat exposure harms growth, employment growth and natural resource rents are strongly positive, and inflation has a negative effect.

5 Local Projections Analysis

5.1 Mathematical Framework and Rationale

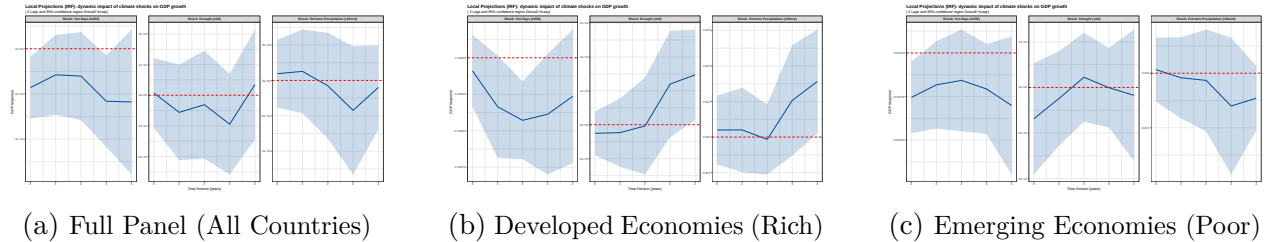
To estimate the dynamic impact of extreme climate shocks on cumulative macroeconomic growth, we employ the Local Projections (LP) framework (Jordà)^{[5][6]}. LP is strictly more robust to dynamic misspecification than traditional VARs and easily accommodates fixed effects without the incidental parameter problem. For each horizon $h = 0, 1, \dots, 4$, we estimate the following sequence of panel regressions:

$$\begin{aligned} \Delta \log(GDP)_{i,t+h} = & \alpha_i^h + \beta_1^h hd30_{i,t} + \beta_2^h cdd_{i,t} + \beta_3^h r20mm_{i,t} \\ & + \sum_{l=1}^p \gamma_l^h \Delta \log(GDP)_{i,t-l} + \sum_{l=0}^p \delta_l^h \mathbf{X}_{i,t-l} + \varepsilon_{i,t+h} \end{aligned} \quad (3)$$

where $\Delta \log(GDP)_{i,t+h}$ is the cumulative real GDP growth for country i , α_i^h represents country fixed effects, and $\mathbf{X}$ contains the differenced macroeconomic controls. We include $p = 2$ lags for all endogenous and exogenous variables to adequately absorb persistent historical dynamics. To correct for the inherent serial correlation of LP at $h > 0$, alongside heteroskedasticity and cross-sectional dependence, we apply the robust Driscoll-Kraay (1998) variance-covariance estimator. Consistent with our previous panel regressions and the theoretical literature, we expect extreme heat ($hd30$) to exert a significant, asymmetric negative drag on output.

5.2 Empirical Results and Discussion

The dynamic impulse response functions (IRFs) are presented in Figure ??, displaying point estimates (β^h) and 95% Driscoll-Kraay confidence intervals.



In the full panel, extreme heat ($hd30$) causes a clear, immediate, and persistent decline in GDP growth, signifying structural scarring rather than a temporary displacement. Responses to precipitation ($r20mm$) and droughts (cdd) remain heterogeneous and statistically weak globally.

Sample stratification confirms pronounced macroeconomic asymmetry. Developed economies exhibit a robust *insulation effect*: adaptive infrastructures and service-oriented markets effectively absorb climate anomalies. Conversely, developing economies suffer severe vulnerability. Their macroeconomic penalty from heat shocks is roughly twice that of rich nations, translating into a permanent structural break with output growth remaining significantly depressed over the entire horizon. This confirms that climate change acts as a regressive multiplier, disproportionately eroding the trajectories of nations lacking adaptive capacity.

6 Panel VAR Analysis

6.1 Mathematical Framework and Identification

To systematically trace the endogenous interactions and indirect transmission channels between climate anomalies and the macroeconomy, we estimate a Panel Vector Autoregression (PVAR)^[9]. The structural

model is defined as:

$$\mathbf{Y}_{i,t} = \mathbf{A}_i + \Phi_1 \mathbf{Y}_{i,t-1} + \Theta \text{time_trend}_{i,t} + \mathbf{u}_{i,t} \quad (4)$$

where $\mathbf{Y}_{i,t}$ is the $k \times 1$ vector of endogenous variables, $\mathbf{A}_i$ captures country-specific fixed effects, and Φ_1 is the matrix of autoregressive coefficients. We incorporate a linear deterministic `time_trend` to capture secular, unobserved global technological progress; this is strictly distinct from the human capital variable (`hc`), as it ensures the long-run stability of the VAR system against global deterministic drifts.

In dynamic panel data, the Fixed Effects estimator is notoriously subject to Nickell (1981) bias. Consequently, the literature frequently employs Generalized Method of Moments (GMM) estimators. However, our dataset constitutes a macro-panel with a large time dimension ($T \approx 63$ years). Because the dynamic panel bias converges to zero at a rate of $1/T$, the bias in our full panel is asymptotically negligible. Thus, we estimate the system via Fixed Effects OLS (PVAR-OLS), completely avoiding the weak instrument proliferation and finite-sample distortions commonly associated with GMM in highly persistent macroeconomic series.

To isolate structural shocks, we orthogonalize the reduced-form residuals using a Cholesky decomposition based on a Wold-causal ordering:

$$\mathbf{Y}_{i,t} = \begin{bmatrix} hd30, & cdd, & r20mm, & d_hc, & d_log_rnn, & d_log_emp, & d_log_GDP, & log_infl \end{bmatrix}'_{i,t} \quad (5)$$

This ordering reflects strict macroeconomic exogeneity: climate anomalies are placed first, as current weather is strictly exogenous to contemporaneous economic activity. Slow-moving stock variables (human and physical capital) follow, while fast-moving flow variables (GDP growth and inflation) are placed last, allowing them to react contemporaneously to all system shocks.

6.2 Lag Selection and Methodological Advantages

Unlike Local Projections (LP), which estimate responses independently at each horizon, the PVAR explicitly maps the entire system of endogenous equations, capturing the complex feedback loops and indirect transmission channels of climate shocks through labor and capital.

To determine the optimal lag structure for the PVAR, we evaluated the standard information criteria on the demeaned system. The results are summarized in Table ??.

Lag (p)	AIC	HQ	Schwarz (SC/BIC)
1	-23.242	-23.176	-23.064
2	-23.364	-23.233	-23.007
3	-23.470	-23.273	-22.935

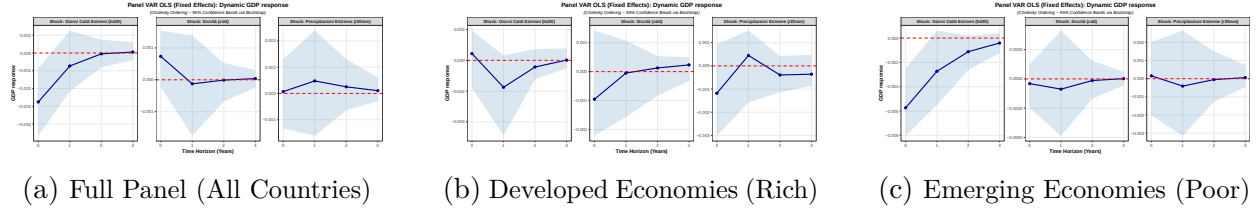
Note: Lowest values indicate the preferred model.

While the Akaike (AIC) and Hannan-Quinn (HQ) criteria suggest a higher-order model ($p = 3$), the Schwarz Criterion (SC/BIC)—which heavily penalizes model complexity—is minimized at $p = 1$. To preserve the degrees of freedom and ensure strict parsimony within our multi-variable dynamic system, we select a lag order of 1.

6.3 Empirical Results and Economic Discussion

We expect the PVAR Orthogonalized Impulse Response Functions (OIRFs) to confirm our hypothesis: extreme heat creates an immediate contractionary shock to GDP. To compute reliable 95% confidence intervals

without Gaussian assumptions, we utilize a block bootstrap procedure. The resulting OIRFs for the full panel and the stratified sub-samples are shown in Figure ??.



The PVAR results elegantly corroborate the findings of the Local Projections, while adding the precision of endogenous feedback. In the full sample (Figure 4a), an orthogonal shock to extreme hot days ($hd30$) triggers an immediate macroeconomic contraction, depressing GDP growth upon impact and reaching its negative peak at year 1 before slowly mean-reverting.

When stratified by economic development, the structural asymmetry becomes glaringly evident. Developed economies (Figure 4b) exhibit a contained initial shock that is rapidly absorbed; their economic systems leverage robust capital accumulation and efficient labor markets to effectively bounce back to equilibrium. Conversely, Developing economies (Figure 4c) suffer a catastrophic initial collapse in output. The shock’s magnitude is significantly deeper, confirming that emerging markets lack the stabilizing endogenous mechanisms (e.g., adaptive physical capital and flexible human capital) required to buffer hydro-meteorological extremes. Ultimately, the PVAR definitively illustrates that climate shocks disrupt the core structural equilibrium of vulnerable nations.

7 ARMA Models

In this section of the project, dedicated to ARMA modeling, we chose not to rely on panel-based evaluations. Given the heterogeneous nature of the dataset, modeling economic growth at the panel level would likely fail to capture the specific dynamics of individual countries.

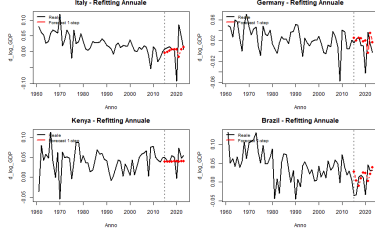
To assess whether an ARMA specification would be appropriate for our analysis, we first examine the autocorrelation structure of the target variable d_log_GDP . Specifically, we analyze the ACF and PACF functions and perform the Ljung–Box test (null hypothesis: no autocorrelation), with the test conducted for lags up to 4, on a sample of countries.

Since the results generally indicate the presence of autocorrelation in several countries, we conclude that ARMA models represent a suitable framework for our analysis.

7.1 Basic ARMA

To estimate the model we divide the data in train set (1960-2014) and test set (2015-2023), then starting from a fit on the train set we decided to implement a recursive window with annual refitting.

The results are not great, in the spirit of clarity we will only show in this report the countries selected as a sample before:



This is, as predictable, especially critical for Kenya since its nature is not autoregressive. In any case, ARMA models in the literature generally fail to deliver satisfactory results. Due to its simplicity, it will function as a useful baseline for more complex models.

To end this analysis rigorously a residual check (1-4 hold, gaussianity does not).

7.2 ARMAX Extension

As exogenous regressors in the ARMAX specification, we use the climate-related variables `hd30`, `cdd`, and `r20mm`.

We first split the dataset into four periods: a training set for the final ARMAX model (1960–2014), a training set for conformal prediction (1960–2000), a calibration set for conformal prediction (2001–2014), and a test set (2015–2023).

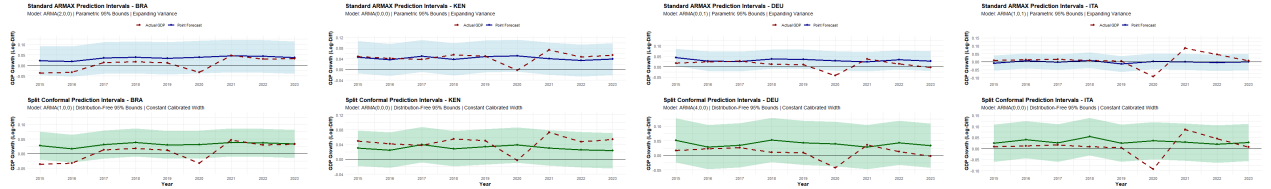
Step 1. Fit the ARMAX model on the training set (1960–2014) and generate forecasts of `d_log_GDP` for the period 2015–2023.

Step 2. Fit an auxiliary ARMAX model using observations up to 2000.

Step 3. Use the auxiliary model to compute the nonconformity scores on the calibration set.

Step 4. Construct prediction intervals using the 95% quantile of the nonconformity scores.

The results for the selected countries are shown below:



(a) Brazil

(b) Kenya

(c) Germany

(d) Italy

ARMAX forecasts with Gaussian prediction intervals and conformal prediction intervals.

Overall, forecasting performance deteriorates during the test period due to the unprecedented economic disruption caused by the COVID-19 crisis.

Moreover, the forecasts for Germany and Kenya are nearly constant over time. This behaviour is consistent with the weak autocorrelation structure observed in these countries. In particular, for Kenya the model selection procedure identifies an ARMAX(0,0,0) specification, resulting in essentially static forecasts. Consistently, the probabilistic evaluation via Pinball Loss demonstrates that the Conformal Prediction framework performs on par with the standard parametric approach, achieving a comparable average loss (0.0079 versus 0.0083) while providing valid coverage without relying on rigid distributional assumptions.

8 Forecasting

To identify the most suitable forecasting model, we compare the different methodologies using the train–test split introduced in the ARMA analysis (1960–2014 | 2015–2023). Model performance is evaluated using three

standard forecasting metrics: Mean Squared Error (MSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE).

Moreover, for consistency, each panel-data methodology is compared with its single-country counterpart (e.g., pVAR is replaced by VAR). An annual rolling-window estimation strategy is employed whenever possible. Under this setting, however, the local projection approach reduces to a standard fixed-effects linear regression and is therefore excluded from the comparison.

Method	N	MSE	MAE	sMAPE	R^2_{OOS}
Mixed Effects (LME)	288	0.0008612	0.01928	61.27	0.40521
Fixed Effects (FE)	288	0.0008499	0.02099	77.81	0.41296
Panel VAR (PVAR)	288	0.0015017	0.02483	73.11	-0.03721
ARMAX Standard	288	0.0018096	0.02714	78.81	-0.24986

From these results we selected Fixed Effect and Mixed Effect for forecasting.

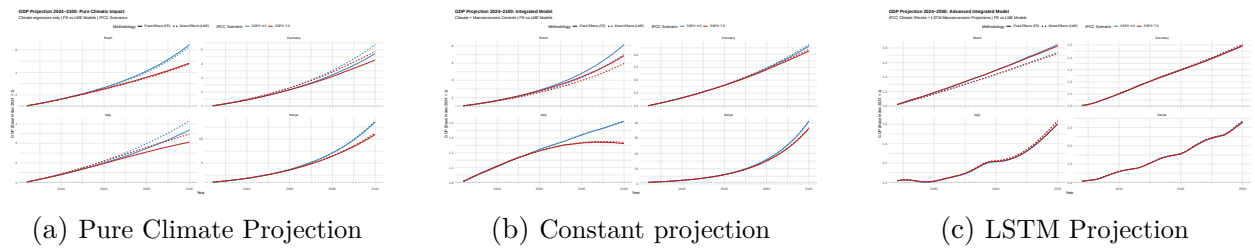
8.1 Forecasting Scenarios

Having selected the best-performing model, we use it to forecast d_log_GDP over the period 2024–2050 under two alternative climate scenarios developed by the IPCC.

- **SSP2–4.5:** a middle-of-the-road scenario characterized by moderate socioeconomic development and climate policies, with no major positive or negative shifts in global cooperation.
- **SSP3–7.0:** a fragmentation scenario characterized by weak international cooperation, slower economic development and continued dependence on fossil fuels.

We performed forecasts under three different assumptions:

- **Pure Climate Projection:** Macroeconomic variables are not included.
- **Constant Projection:** Macroeconomic variables are held constant at their last observed values (2023).
- **LSTM Projection:** Macroeconomic variables are forecasted using an LSTM model.



9 Excluded models

In our analysis we excluded cointegration (variables of interest were $I(0)$), ARCH and GARCH (There is no conditional volatility)

Bibliography

- [1] Dell, M., Jones, B. F., & Olken, B. A. (2012). *Temperature shocks and economic growth: Evidence from the last half-century*. American Economic Journal: Macroeconomics, 4(3), 66–95.
- [2] Burke, M., Hsiang, S. M., & Miguel, E. (2015). *Global non-linear effect of temperature on economic production*. Nature, 527(7577), 235–239.
- [3] Paglialunga, E., Coveri, A., & Zanfei, A. (2022). *Climate change and within-country inequality: New evidence from a global perspective*. World Development, 159, 106031.
- [4] Im, K. S., Pesaran, M. H., & Shin, Y. (2003). *Testing for unit roots in heterogeneous panels*. Journal of Econometrics, 115(1), 53–74.
- [5] Jordà, Ó. (2005). *Estimation and inference of impulse responses by local projections*. American Economic Review, 95(1), 161–182.
- [6] Jordà, Ó., & Taylor, A. M. (2025). *Local Projections for Macroeconometrics*. Journal of Economic Literature, (Forthcoming/In Press).
- [7] Montiel Olea, J. L., & Plagborg-Møller, M. (2026). *Simultaneous identification and causal ordering in structural macroeconometrics*. Econometrica, (Forthcoming/In Press). ordering applied to physical and economic variables.
- [8] Kanzig, D. R. (2021). *The macroeconomic effects of carbon extremes and climate anomalies*. Working Paper.
- [9] Sigmund, M., & Ferstl, R. (2021). *Panel Vector Autoregression in R with the Package panelvar*. Journal of Statistical Software, 97(5), 1–32.